

# YOUNGJAE MIN (민영제)

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| RESEARCH INTERESTS | Reliable AI & Autonomy, Learning-Based Control, Continual Learning, Optimization                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                    | I develop learning-enabled control methods for reliable intelligent systems that are safe, adaptive, and deployable in safety-critical settings. My work integrates machine learning, control, and optimization to design principled algorithms with theoretical guarantees and practical reliability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| EDUCATION          | <div><div><b>Massachusetts Institute of Technology</b> June 2023 - (Dec. 2026)<br/>Ph.D. Candidate in Autonomous Systems and Statistics Advisor: Professor <a href="#">Navid Azizan</a><ul style="list-style-type: none"><li>Major: Machine Learning / Minor: Control</li></ul></div><div><b>Massachusetts Institute of Technology</b> Sep. 2021 - May 2023<br/>S.M. in Aeronautics and Astronautics</div><div><b>Korea Advanced Institute of Science and Technology</b> Mar. 2014 - Feb. 2020*<br/>B.S. in Electrical Engineering, Mathematical Sciences (<i>summa cum laude</i>) (*two years in military)</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| EXPERIENCE         | <div><div><b>Laboratory for Information and Decision Systems, MIT</b> Sep. 2021 - (Dec. 2026)<br/><i>Research Assistant / Advisor: Prof. Navid Azizan</i><ul style="list-style-type: none"><li>Learning and control with provable input-output constraint satisfaction [P1, P3, C7]</li><li>Online/continual learning robust to catastrophic forgetting [P4, C6, C9, J5]</li></ul></div><div><b>Autopilot Team, Tesla, CA</b> May 2026 - Aug. 2026<br/><i>Intern / Advisor: Dhiral Chheda</i><ul style="list-style-type: none"><li>Motion planning for Tesla's Autopilot software</li></ul></div><div><b>CI Mobility Research Team, Honda Research Institute, CA</b> May 2025 - Aug. 2025<br/><i>Research Intern / Advisor: Sangjae Bae</i><ul style="list-style-type: none"><li>Motion planning via integrated offline RL and online planning [P2]</li></ul></div><div><b>Laboratory for Information and Control Systems, KAIST</b> Feb. 2019 - July 2021<br/><i>Undergrad. Researcher / Advisor: Prof. Han-Lim Choi</i><ul style="list-style-type: none"><li>Real-time 3-D mapping of dynamic environments [C8, C11]</li><li>Learning and planning of dynamical systems [C12, J10, J14]</li></ul></div><div><b>Inference and Information for Data Science Lab, KAIST</b> Mar. 2018 - Jan. 2019<br/><i>Undergrad. Researcher / Advisor: Prof. Hye Won Chung</i><ul style="list-style-type: none"><li>Representational capability of neural networks [C13]</li></ul></div><div><b>Signal Kinetics Group, Media Lab, MIT</b> June 2018 - Sep. 2018<br/><i>Visiting Researcher / Advisor: Prof. Fadel Adib</i><ul style="list-style-type: none"><li>Non-contact vital sign monitoring via mmWave radar</li></ul></div><div><b>Networking &amp; Mobile Systems Lab, KAIST</b> June 2017 - Feb. 2018<br/><i>Undergrad. Researcher / Advisor: Prof. Sung-Ju Lee</i><ul style="list-style-type: none"><li>Indoor person localization via Wi-Fi signals</li></ul></div></div> |
| PUBLICATIONS       | (P: preprint, C: conference proceedings, J: journal articles, *equally contributed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                    | [P1] “HardNet: Hard-Constrained Neural Networks with Universal Approximation Guarantees,” <i>preprint</i><br><b>Youngjae Min</b> , Navid Azizan <a href="#">[arXiv: 2410.10807]</a><br>NeurIPS Constrained ML Workshop 2025 (selected as Oral)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

- [P2] “HOLO-MPPI: Multi-Scenario Motion Planning via Hierarchical Policy Optimization,” *preprint*  
**Youngjae Min**, Jovin D’sa, Faizan M Tariq, David Isele, Navid Azizan, Sangjae Bae
- [P3] “HardNet++: Nonlinear Constraint Enforcement in Neural Networks,” *preprint*  
 Andrea Goertzen, Kaveh Alim, **Youngjae Min**, Navid Azizan [\[arXiv: 2604.19669\]](#)
- [P4] “ORFit: Efficient One-Pass Learning for Overparameterized Models,” *preprint*  
**Youngjae Min**, Namhoon Cho, Navid Azizan
- [J5] “SketchOGD: Memory-Efficient Continual Learning”  
**Youngjae Min**, Benjamin Wright, Jeremy Bernstein, Navid Azizan  
**TSP 2026 - IEEE Transactions on Signal Processing** [\[arXiv: 2305.16424\]](#)
- [C6] “II-ORFit: One-Pass Learning with Bregman Projection”  
 Namhoon Cho, **Youngjae Min**, Hyo-Sang Shin, Navid Azizan  
**ACC 2024 (Invited Session) - American Control Conference**
- [C7] “Data-Driven Control with Inherent Lyapunov Stability”  
**Youngjae Min**, Spencer M. Richards, Navid Azizan  
**CDC 2023 (Invited Session) - IEEE Conference on Decision and Control** [\[arXiv: 2303.03157\]](#)
- [C8] “DS-K3DOM: 3-D Dynamic Occupancy Mapping with Kernel Inference and Dempster-Shafer Evidential Theory,” Juyeop Han\*, **Youngjae Min\***, Hyeok-Joo Chae, Byeong-Min Jeong, Han-Lim Choi  
**ICRA 2023 - IEEE International Conference on Robotics and Automation** [\[arXiv: 2209.07764\]](#)
- [C9] “One-Pass Learning via Bridging Orthogonal Gradient Descent and Recursive Least-Squares”  
**Youngjae Min**, Kwangjun Ahn, Navid Azizan  
**CDC 2022 (Invited Session) - IEEE Conference on Decision and Control** [\[arXiv: 2207.13853\]](#)
- [J10] “Online Gaussian Process State-Space Model: Learning and Planning for Partially Observable Dynamical Systems,” Soon-Seo Park, Young-Jin Park, **Youngjae Min**, Han-Lim Choi  
**IJCAS 2022 - International Journal of Control, Automation and Systems** [\[arXiv: 1903.08643\]](#)
- [C11] “Kernel-Based 3-D Dynamic Occupancy Mapping with Particle Tracking”  
**Youngjae Min**, Do-Un Kim, Han-Lim Choi  
**ICRA 2021 - IEEE International Conference on Robotics and Automation**
- [C12] “Informative Planning of Mobile Sensor Networks in GPS-Denied Environments”  
**Youngjae Min**, Soon-Seo Park, Han-Lim Choi  
**SciTech 2020 - AIAA Science and Technology Forum and Exposition** [\[arXiv: 1909.11046\]](#)
- [C13] “Shallow Neural Network can Perfectly Classify an Object following Separable Probability Distribution”  
**Youngjae Min**, Hye Won Chung  
**ISIT 2019 - IEEE International Symposium on Information Theory** [\[arXiv: 1904.09109\]](#)
- [J14] “A Distributed ADMM Approach to Non-Myopic Path Planning for Multi-Target Tracking”  
 Soon-Seo Park\*, **Youngjae Min\***, Jung-Su Ha, Doo-Hyun Cho, Han-Lim Choi  
**Access 2019 - IEEE Access** [\[arXiv: 1807.11068\]](#)

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|--------------------|-------------------------------------------------------------------------------------|-------------|
| HONORS<br>& AWARDS | MIT SoE MathWorks Fellowship, <i>MIT School of Engineering &amp; MathWorks Inc.</i> | 2025 - 2026 |
|                    | KEF Scholarship, <i>Kwanjeong Educational Foundation</i>                            | 2023 - 2027 |
|                    | Global Leadership Award, <i>KAIST</i>                                               | 2020        |
|                    | ◦ Awarded to the top undergraduate for outstanding creativity                       |             |
|                    | GE Foundation Scholar-Leaders Program, <i>Fulbright / GE Foundation</i>             | 2015 - 2019 |

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|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
|                            | KFAS Undergraduate Student Scholarship, <i>Korea Foundation for Advanced Studies</i>                                                                                                                                                                                                                                                                                                               | 2015 - 2019                       |
|                            | KAIST Presidential Fellowship, <i>KAIST</i>                                                                                                                                                                                                                                                                                                                                                        | 2014                              |
|                            | ◦ Awarded to the top 11 students of the Class of 2018                                                                                                                                                                                                                                                                                                                                              |                                   |
| PROFESSIONAL<br>ACTIVITIES | <b>Reviewing</b> : American Control Conference (ACC), IEEE Conference on Decision and Control (CDC), IEEE Control Systems Letters (L-CSS), IEEE International Conference on Robotics and Automation (ICRA), IEEE Robotics and Automation Letters (RA-L), IEEE Transactions on Robotics (T-RO), International Conference on Learning Representations (ICLR), Learning for Dynamics & Control (L4DC) |                                   |
| TEACHING<br>EXPERIENCE     | 2.004 Dynamics and Control II, Teaching Assistant, <i>MIT</i>                                                                                                                                                                                                                                                                                                                                      | Fall'22                           |
|                            | PH141 General Physics I, Tutor, <i>KAIST</i>                                                                                                                                                                                                                                                                                                                                                       | Spring'15, Spring'17              |
|                            | KAIST Global Institute for Talented Education, Online Tutor, <i>KAIST</i>                                                                                                                                                                                                                                                                                                                          | Aug. 2014 - July 2015             |
| EXTRACURRI-<br>CULARS      | Charles River Running Crew, President, <i>MIT</i>                                                                                                                                                                                                                                                                                                                                                  | May 2024 - May 2025               |
|                            | KAIST EE Newsletter, Reporter, <i>KAIST</i>                                                                                                                                                                                                                                                                                                                                                        | Mar. 2017 - Dec. 2017             |
|                            | ROK Army & U.S. Army, IT Specialist (Sergeant, KATUSA), <i>Cp. Carroll</i>                                                                                                                                                                                                                                                                                                                         | July 2015 - Apr. 2017             |
|                            | The Real LUNATIC, B-Boy, <i>KAIST</i>                                                                                                                                                                                                                                                                                                                                                              | Mar. 2014 - July 2015             |
| SKILLS                     | <b>Programming</b>                                                                                                                                                                                                                                                                                                                                                                                 | Python, C/C++, CUDA               |
|                            | <b>Frameworks &amp; Tools</b>                                                                                                                                                                                                                                                                                                                                                                      | PyTorch, ROS, Git                 |
|                            | <b>Languages</b>                                                                                                                                                                                                                                                                                                                                                                                   | Korean (native), English (fluent) |